

- Every Node is either RED or BLACK (Coloring Property)
- Each NULL pointer is Considered to be a BLACK Node
- If a node is RED, then both of its Children are BLACK (RED Node Property)
- Every path from a Node to a Leaf Node contain the same number of BLACK Nodes (BLACK Height Property)
- Root Node is BLACK (ROOT Node Property)



RED BLACK TREES

The main benefit of a RED - BLACK Tree is that it provides a good worst case runtime for its operations

- It is a Self Balancing Tree
- The Height of the tree remains logarithmic in the number of nodes
- Ensure that the Insert / Delete / Search operations run in $O(\log n)$ time where n is the number of nodes in the tree
- Operations are similar to those of BST but with additional steps to adjust colors and perform rotations to maintain the tree's Balancing properties



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Insert Operation

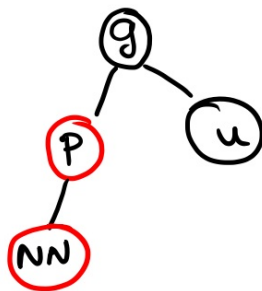
- Create a NewNode (NN) and Color it **RED**
- Insert into RBT using BST Principle
- While Color(Parent(NN)) is **RED** do
 - if Color(Uncle(NN)) is **RED** then
 - Color(Parent(NN)) = Color(Uncle(NN)) = BLACK
 - Color(GrandParent(NN)) = **RED**
 - GrandParent(NN) is now treated as NN
 - else
 - // if the Color(Uncle(NN)) is BLACK, then perform rotations to balance the tree
 - NEXT SLIDE
- endif
- End While
- Color(Root Node) = BLACK



RED BLACK TREES

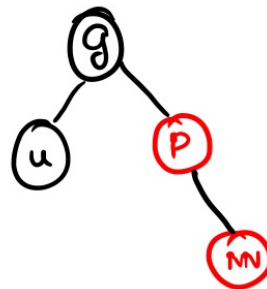
if $\text{color}(\text{Uncle}(\text{NN}))$ is BLACK then any one of the following 4 Scenarios arise

LEFT-LEFT Case



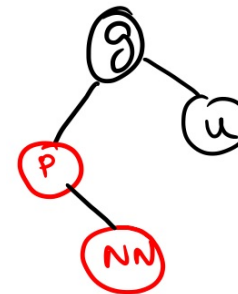
Swap Colors of p and g
Right Rotate on g

RIGHT-RIGHT Case



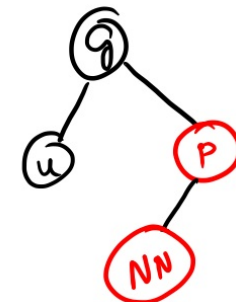
Swap Colors of p and g
Left Rotate on g

LEFT- RIGHT Case



Left Rotate on p
Swap Colors of NN and g
Right Rotate on g

RIGHT- LEFT Case



Right Rotate on p
Swap Colors of NN and g
Left Rotate on g

A Right Rotation around a given Node X is as follows

- Identify the Left child of X as Y
- Make the RST of Y as LST of X
- Elevate Y to the Position where X was originally
- Set X as the Right Child of Y

A Left Rotation around a given Node X is as follows

- Identify the Right child of X as Y
- Make the LST of Y as RST of X
- Elevate Y to the Position where X was originally
- Set X as the Left Child of Y



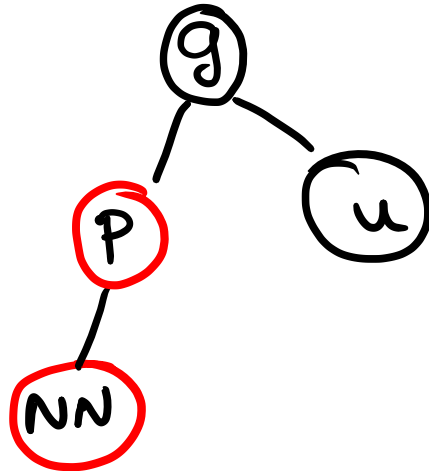
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RED BLACK TREES

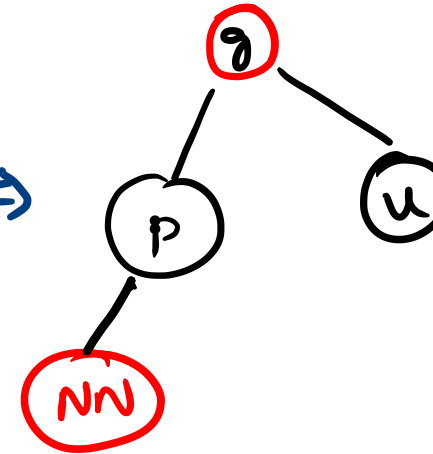
Case 1



* Swap colors
b/w g and P

* RR on g

⇒

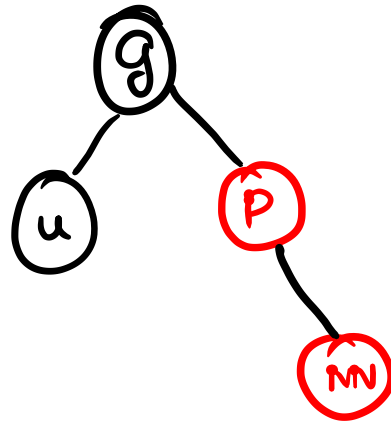




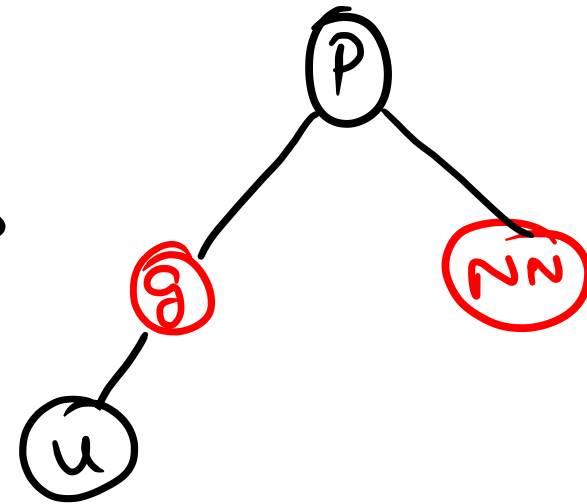
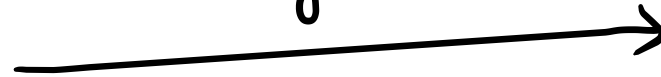
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RED BLACK TREES

Case 2



Swap colors of p, g
LL on g

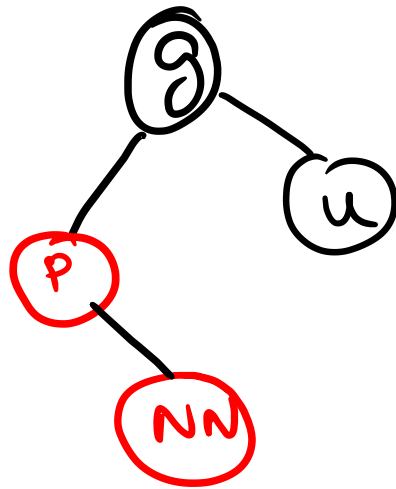




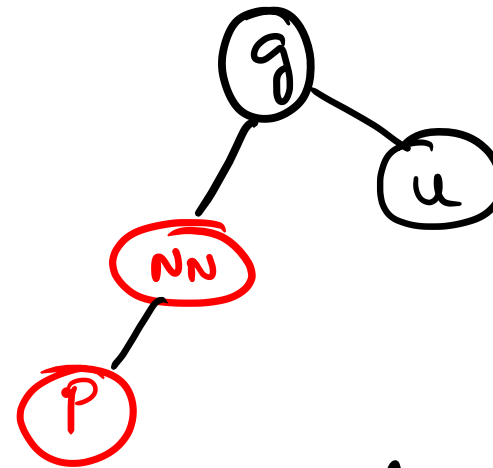
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RED BLACK TREES

Case 3



LL on P

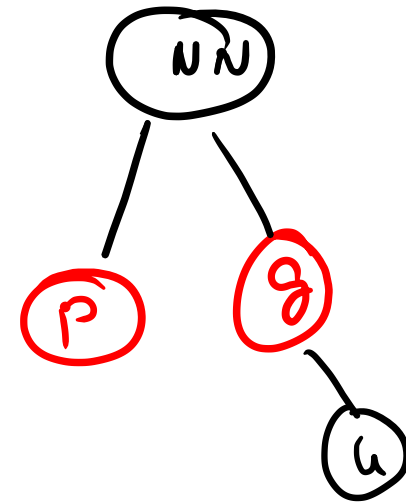
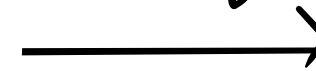


Apply Case 1

+ Swap Colors of
g, NN

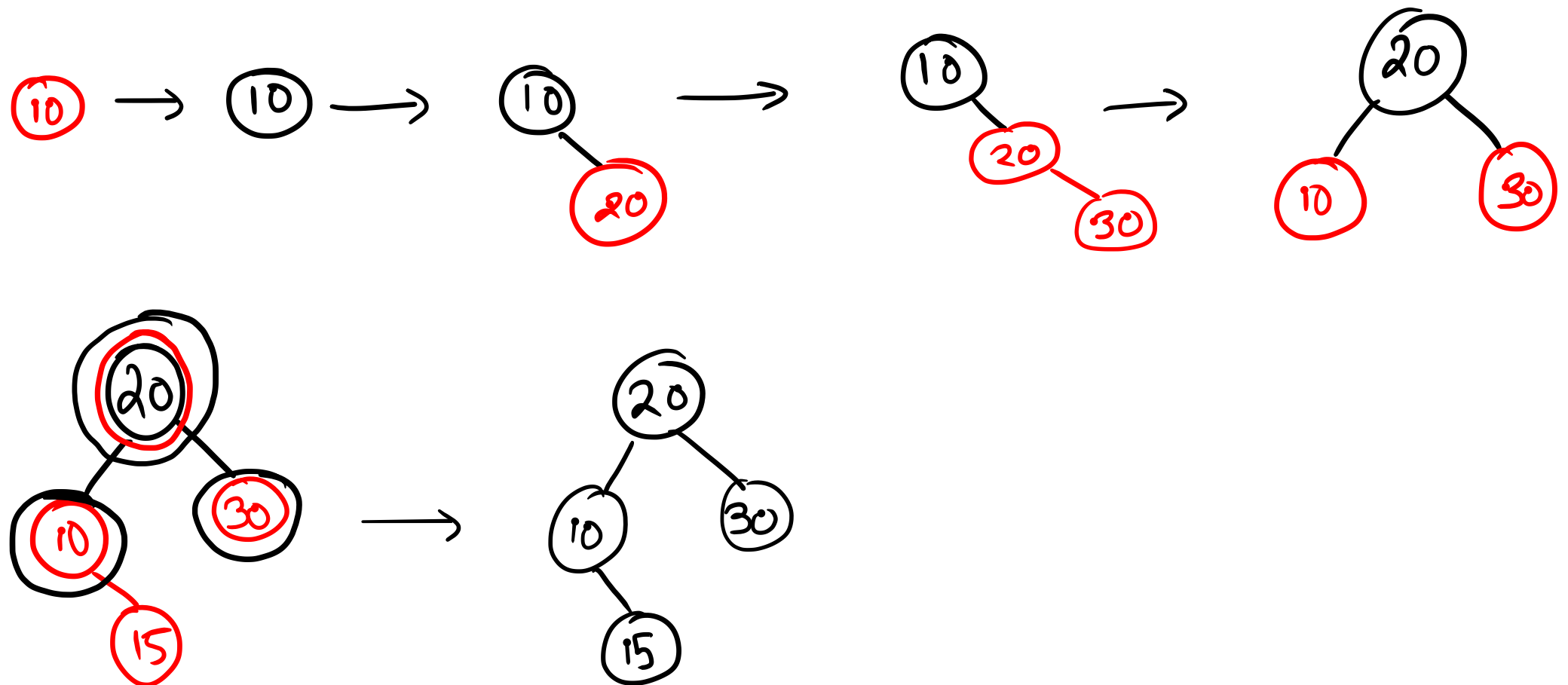
RR on g

Color not on
Red



RED BLACK TREES

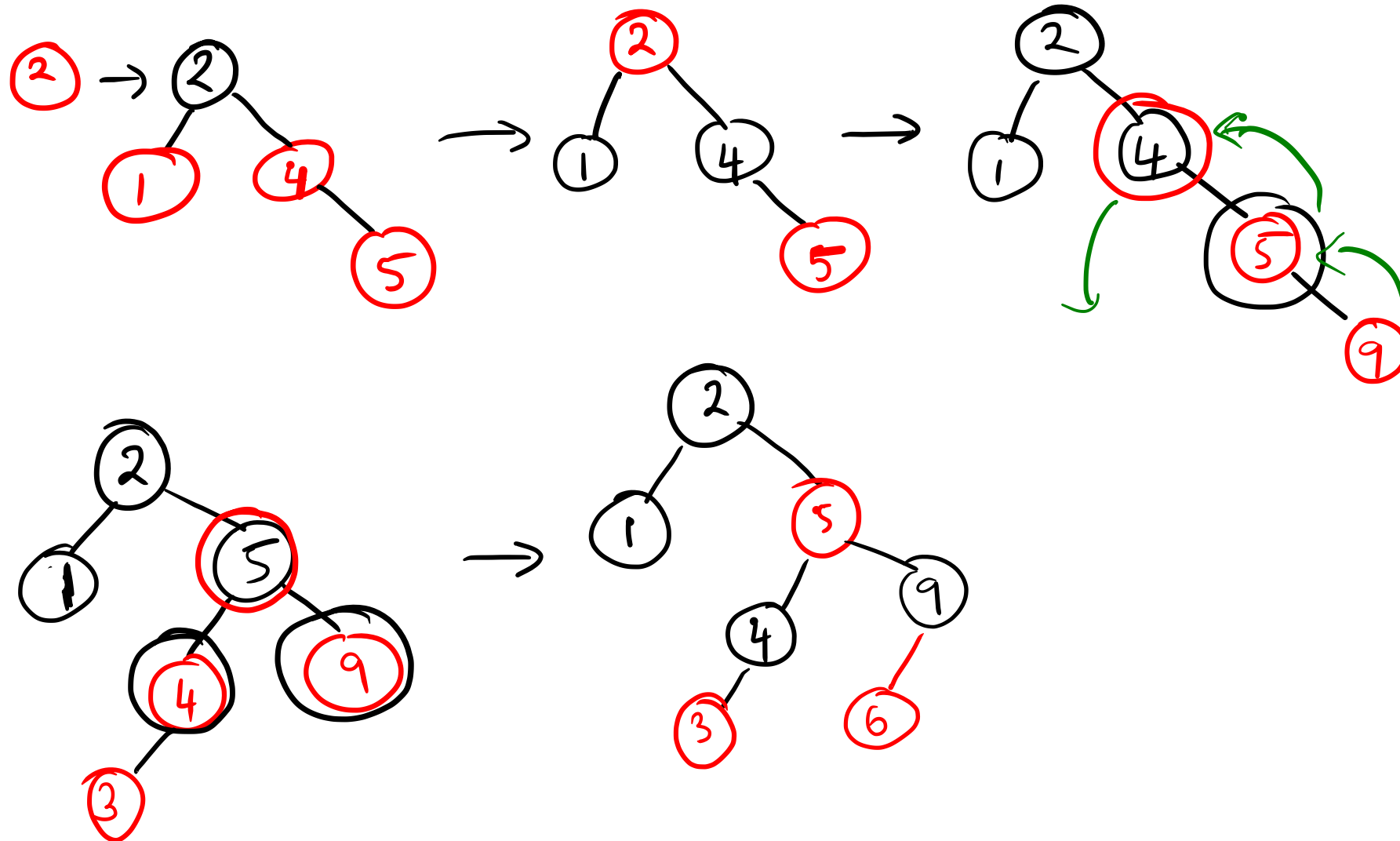
10, 20, 30, 15





2 1 4 5 9 3 6 7

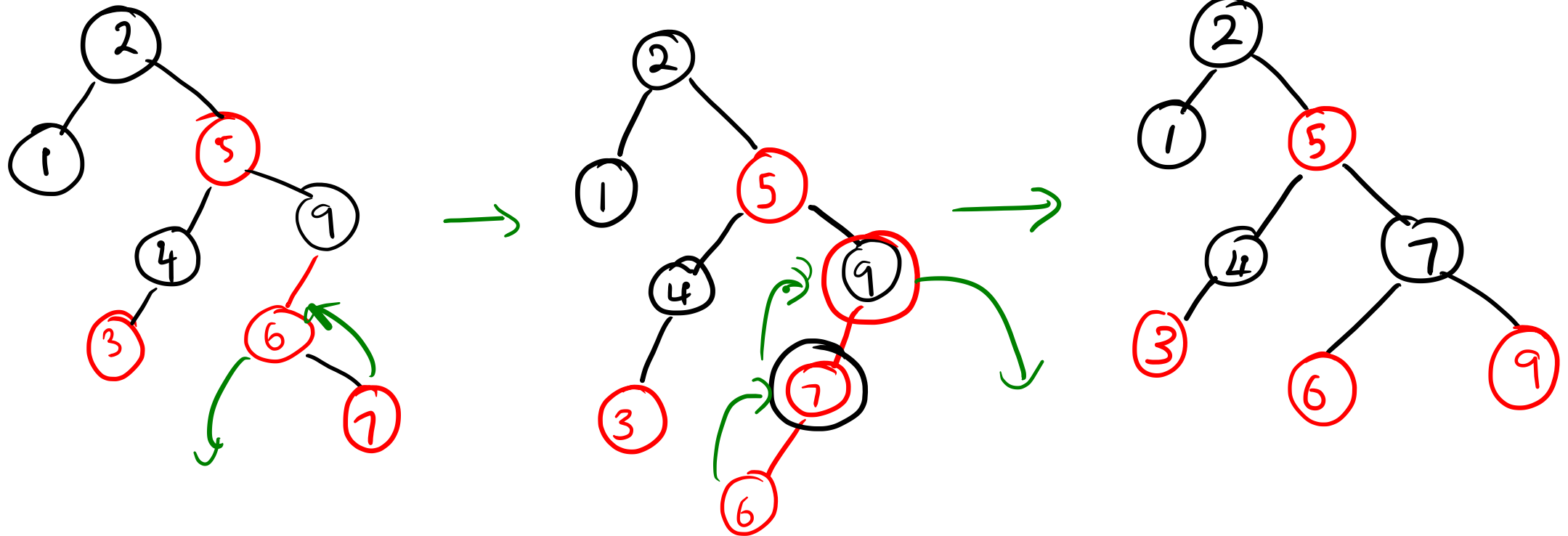
RED BLACK TREES



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RED BLACK TREES

2 1 4 5 9 3 6 7

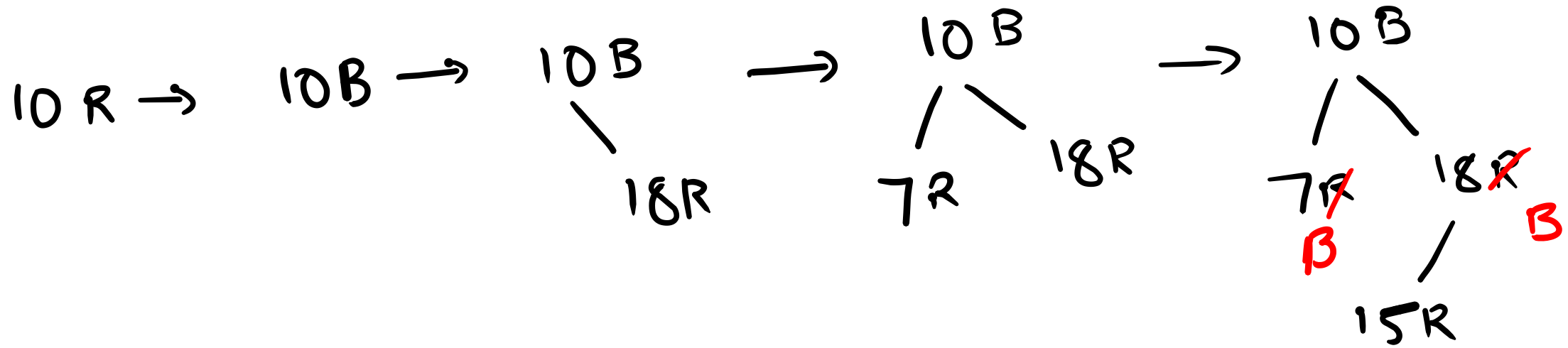




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RED BLACK TREES

✓✓✓✓
10, 18, 7, 15, 16, 30, 25, 40, 60, 2, 1, 70



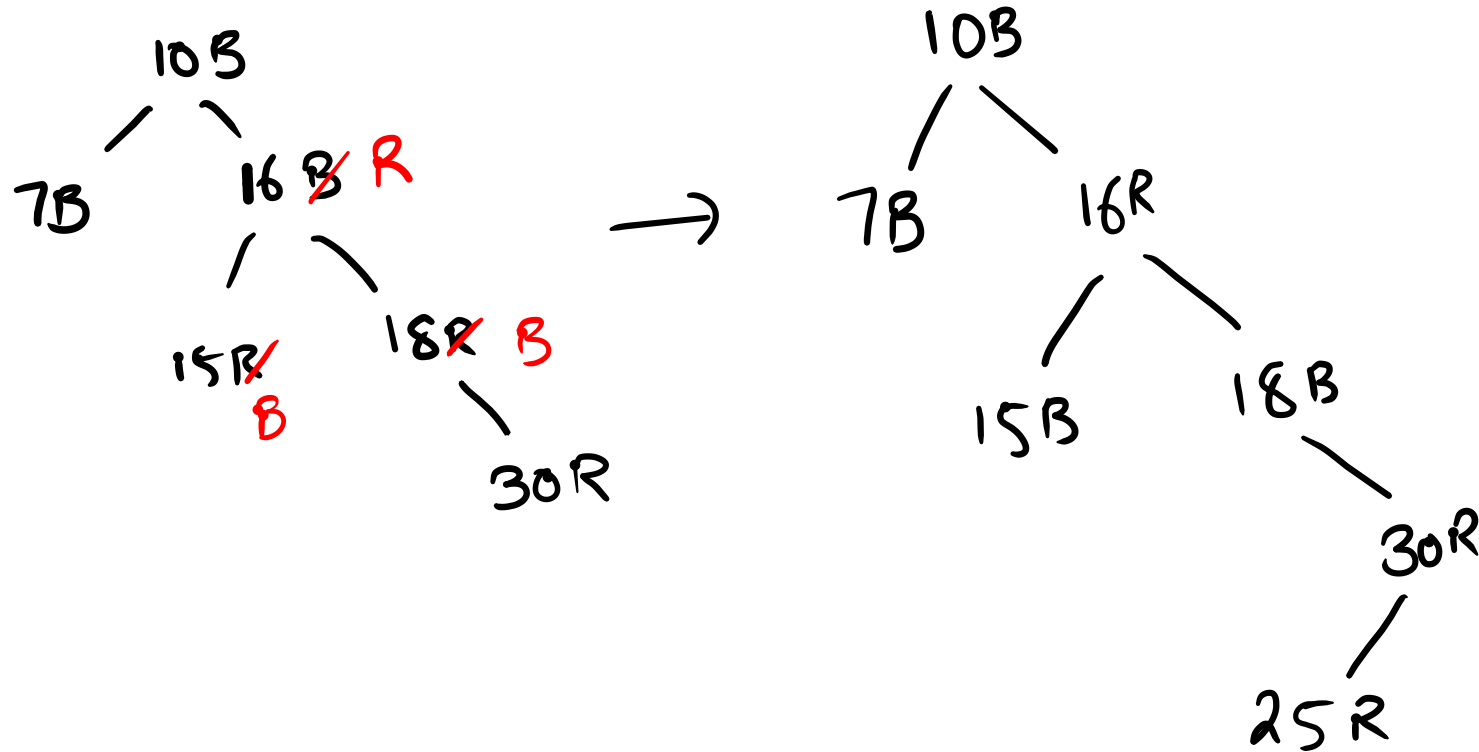




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RED BLACK TREES

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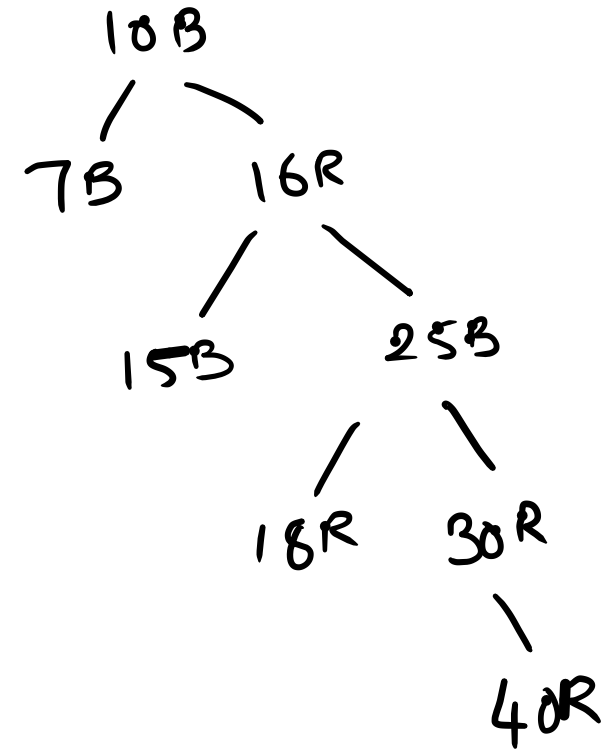
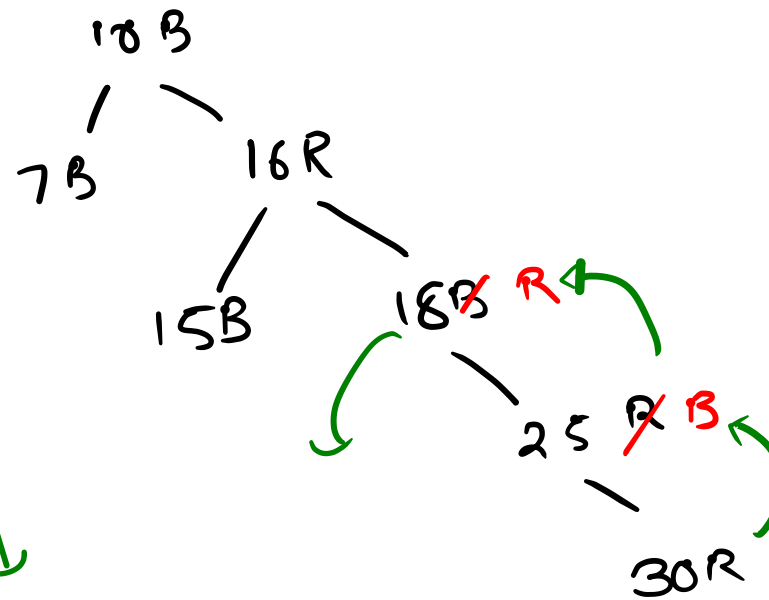
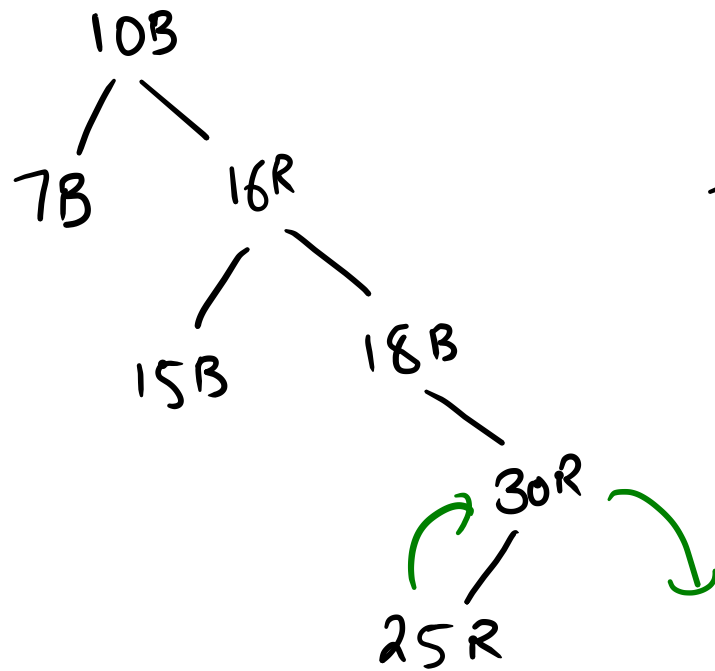




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RED BLACK TREES

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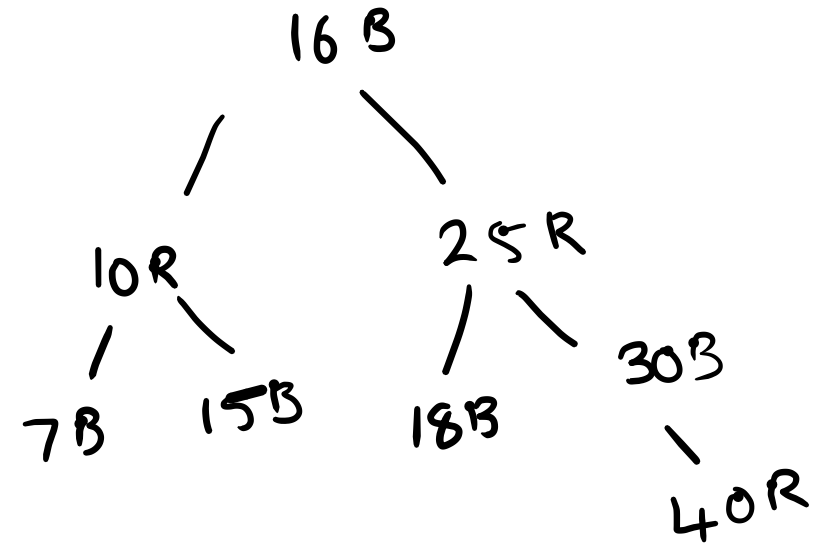
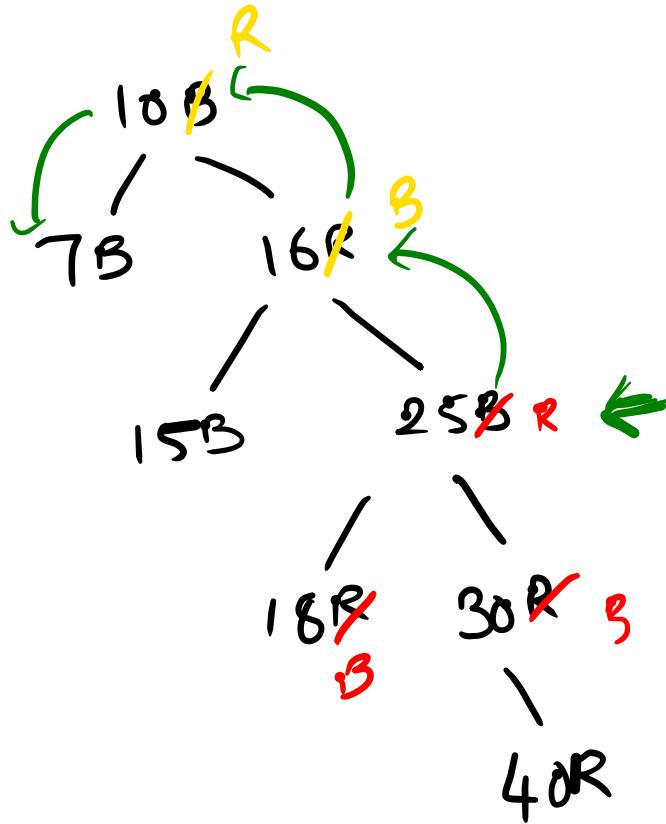




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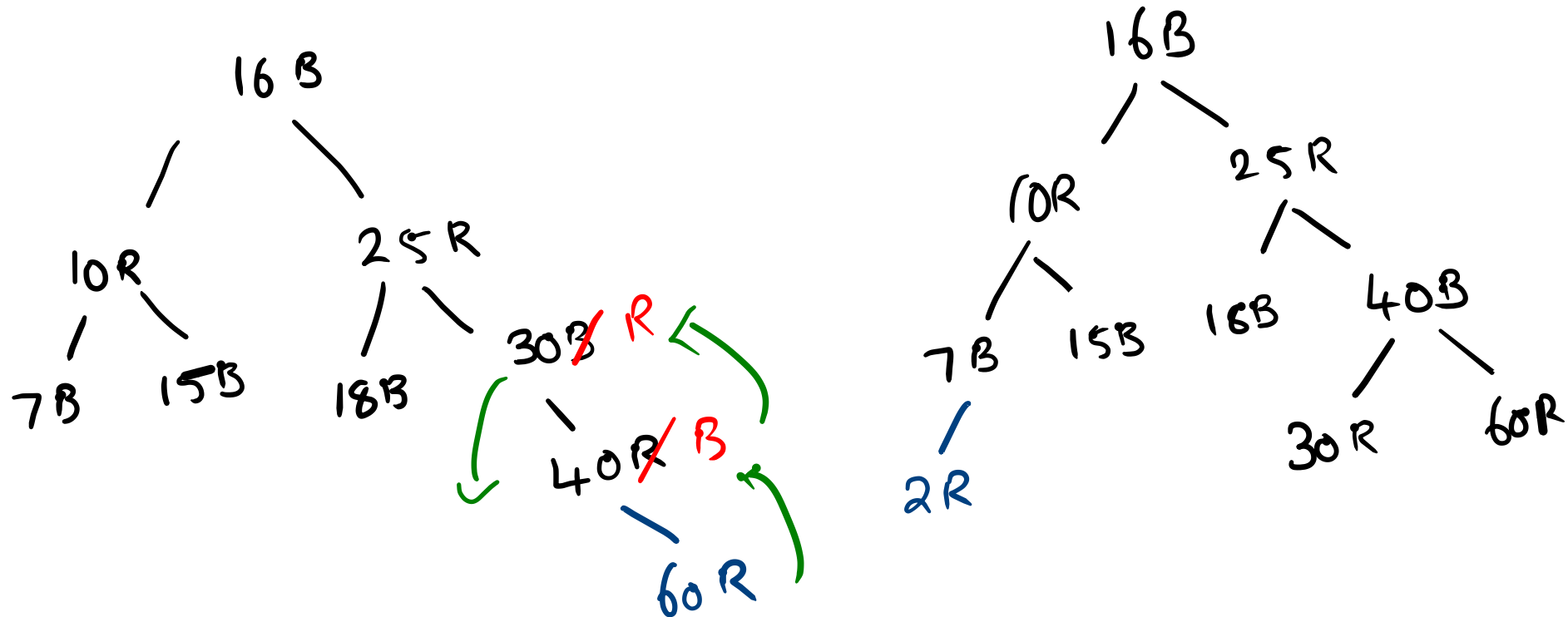
✓ ✓ ✓ ✓ ✓ ✓ 10, 18, 7, 15, 16, 30, 25, 40, 60, 2, 1, 70





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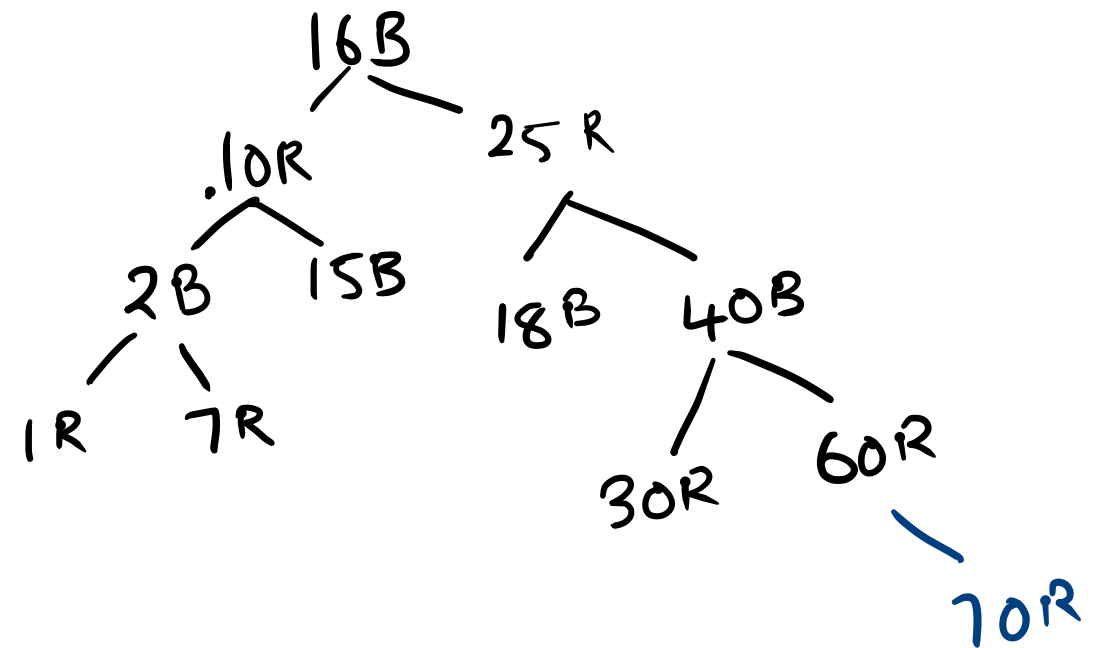
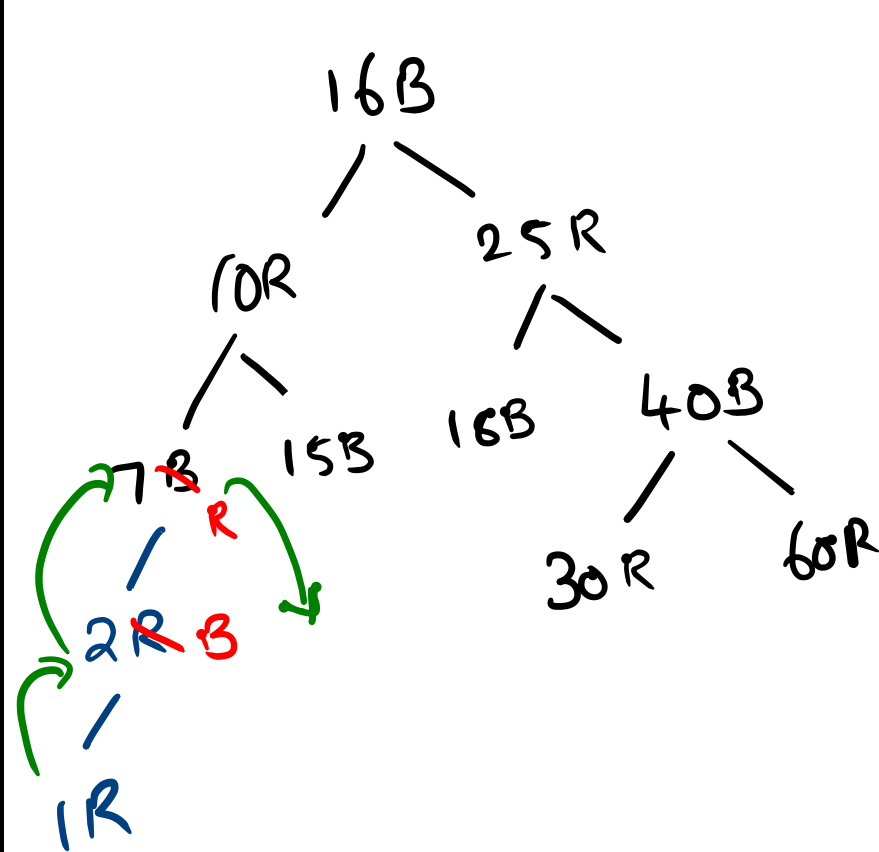
✓ ✓ ✓ ✓ ✓ ✓ RED BLACK TREES ✓
10, 18, 7, 15, 16, 30, 25, 40, 60, 2, 1, 70





✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓
10, 18, 7, 15, 16, 30, 25, 40, 60, 2, 1, 70

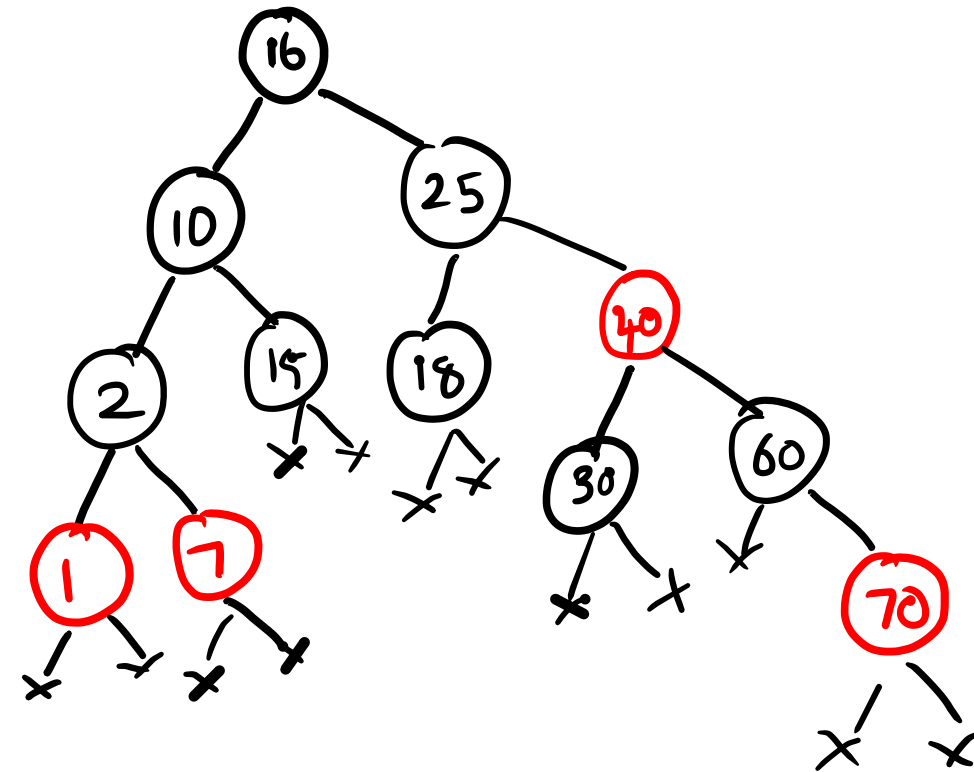
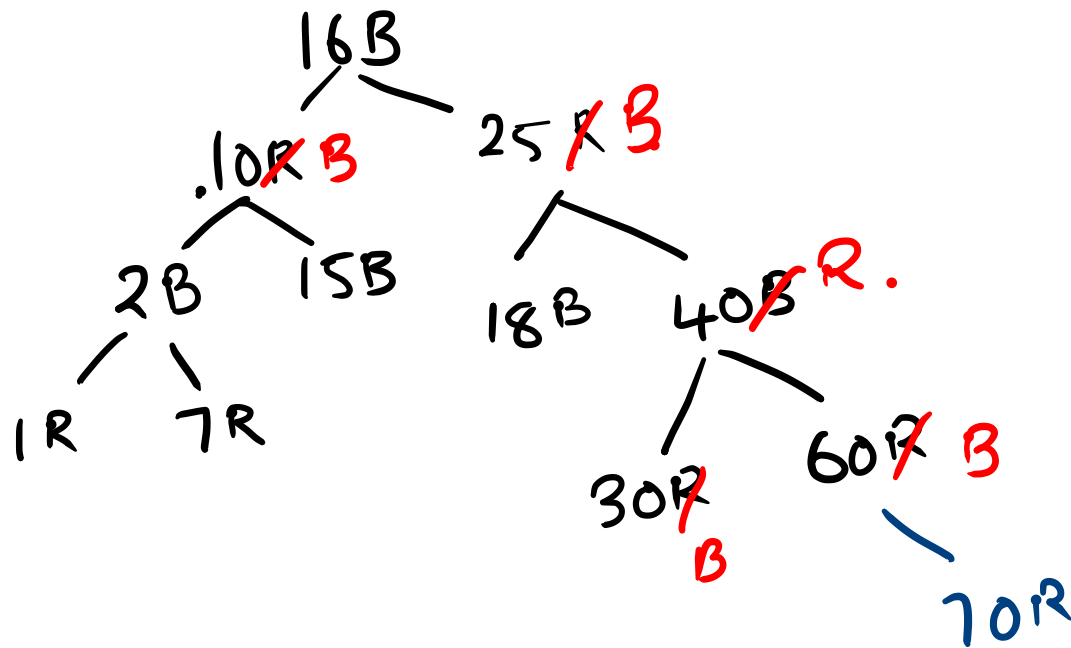
RED BLACK TREES



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✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓
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